

Thermal sensitivity and the role of behavior in driving an intertidal predator–prey interaction

CRISTIÁN J. MONACO,^{1,3} DAVID S. WETHEY,¹ AND BRIAN HELMUTH,²

¹Department of Biological Sciences, University of South Carolina, Columbia, South Carolina 29208 USA

²Marine Science Center, Northeastern University, Nahant, Massachusetts 01908 USA

Abstract. Environmental stress models (ESM) provide a useful framework to study the direct and indirect ecological drivers of community diversity and resilience. ESMs make predictions about the relative importance of structuring processes (e.g., predation) based on the relative stress suffered by consumers and prey. Their practical application, i.e., determining the conditions under which consumers and prey performance is more negatively affected, has been limited because the roles of behavior and physiology are not usually considered. We examined the role of thermal sensitivity and behavior on the thermal performance of the rocky intertidal predator *Pisaster ochraceus* and its main prey *Mytilus californianus*. We propose a novel framework that merges thermal performance curves (TPC) with observations of microhabitat use to provide a realistic perspective of the relative physiological conditions of predator and prey. First, by deriving aquatic and aerial TPCs for both species and from two sites, we found differences in parameter values that in some cases correspond to the individuals' origins. Second, we calculated realized thermal performance in the field by combining TPCs with body temperatures recorded with biomimetic sensors. Notably, thermal performance of *Pisaster* was higher than that for *Mytilus* (i.e., prey-stress model), contrary to previous expectations based on caging experiments. Third, these estimates of thermal performance corresponded loosely with a measured indicator of overall physiological condition (body mass index, BMI) and a marker for extreme thermal stress (heat-shock proteins 70 kDa), suggesting that environmental drivers other than temperature, such as food supply, must be considered. We found no evidence that *Pisaster* movement significantly influences thermal performance under typical conditions, suggesting instead that its preference for sheltered microhabitats provides a mechanism for avoiding exposure to extreme environmental conditions. Through the application of TPCs and ESMs, this study provides a unique perspective on the importance of physiology and behavior in driving the sensitivity of species interactions to environmental change. Crucially, this framework allowed clarifying that this system behaves as a prey- instead of consumer-stress model, which may also apply to many other ectotherm species interactions.

Key words: abiotic drivers; ecological interactions; intertidal; keystone predator; microhabitat; mussels; prey; sea stars; temperature; thermal stress.

INTRODUCTION

Predicting the dynamics of natural systems as a function of environmental drivers requires a mechanistic understanding of the biotic and abiotic factors controlling individual-level processes (Tomanek and Helmuth 2002, Denny and Helmuth 2009). This task is particularly timely given the increasing threats posed by climate change on ecosystems globally, which are expected to present assemblages with novel suites of environmental conditions (Fitzpatrick and Hargrove 2009, Singer et al. 2016). To this end, significant efforts have been oriented toward modeling the independent and interactive effects of multiple environmental stressors

(Gunderson et al. 2016). However, climate change is predicted to impact populations via both direct abiotic (e.g., temperature) and indirect biotic (e.g., ecological interactions) stressors with the potential for seemingly unpredictable synergistic, antagonistic, or additive outcomes (Wernberg et al. 2011, Williams et al. 2011). It has been argued that such outcomes may result from counterintuitive interactions between the different drivers acting on individual organisms (direct physiological effects) or via differential susceptibility to stress among interacting species (indirect effects; Wethey 1983; Crain et al., 2008), including altered patterns of behavior such as foraging rates of predators (Sanford 1999). For example, the direct effects of increasingly warmer temperatures may indirectly force phenological mismatches between key interacting species, thereby disrupting community dynamics (Ohlberger et al. 2014).

Environmental stress models (ESM), a useful framework to anticipate the consequences of ecological

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³Present address: Department of Zoology and Entomology, Rhodes University, Grahamstown, 6140 South Africa. E-mail: cristianmonaco@gmail.com

interactions along gradients of environmental drivers (Menge and Sutherland 1987, Menge and Olson 1990, Menge et al. 2002), can be used as heuristic tools for untangling direct from indirect effects. The theory behind ESMs has seen promising advances over the last 10–20 yr. Importantly, in addition to considering interactions such as predation and competition, ecologists have acknowledged the importance of positive interactions (e.g., facilitation) in driving natural systems' dynamics, especially under climate change (Bertness and Leonard 1997, Leonard 2000, Buckley 2013), and efforts to conceptually include them into the ESM framework have arisen (Bruno et al. 2003, Crain and Bertness 2006).

A central shortcoming of most empirical and theoretical studies applying the ESM framework, however, is the assumption that “environmental stress” is an attribute independent of the organisms that it impacts, and that “stress” thus can be measured in terms of simple gradients instead of how it affects organisms (Wingfield and Kitaysky 2002, Kearney 2006). Such assumptions thus often lack the ecological realism that is necessary to accurately characterize context dependency, but which can potentially explain the observation that the strength and direction of interspecific interactions varies among environments (Crain 2008). In particular, many studies (but see, e.g., Menge et al. 2007) have been unable to incorporate interspecific variability in physiological sensitivity and behavioral responses (e.g., microhabitat choice) to environmental drivers, despite acknowledging their importance (Petes et al. 2008). Because ecological interactions, microhabitat use, and physiological responses are tightly interdependent in many ecosystems (Porter et al. 1975, Dahlhoff et al. 2001), studies would benefit by considering them in concert (Monaco and Helmuth 2011).

When applied to predator–prey systems, ESMs have taken one of two forms: consumer-stress models (CSM) or prey-stress models (PSM). The system behaves as a CSM if predators appear more negatively affected by the environment than their prey (Menge and Sutherland 1976); alternatively, if prey suffer more from environmental stressors, the system is labeled as a PSM (Trowbridge 1998, Menge et al. 2002). Depending on whether we use CSMs or PSMs, predictions about the dynamics of interacting species may follow fundamentally different trajectories (Menge et al. 2002). Thus, for ESMs to serve their purpose, it is essential to accurately identify which model is likely to apply to each particular study system, and between each pair of interacting species (Crain 2008).

The influence of temperature on species interactions can be appreciated in every type of habitat, including terrestrial, freshwater, and marine (Dell et al. 2011, 2013). For example, the influence of temperature on the foraging schedules of insects (Jayatilaka et al. 2011) and lizards (Pianka 1973) is well documented. Notably, because of its steep physical and biological gradients, marine intertidal systems have long served as a natural

laboratory to develop and test ESMs (Menge and Sutherland 1987, Scrosati et al. 2011). With the constant rise and fall of tides, intertidal organisms frequently cope with multiple physical stressors such as extremes in low- and high-tide body temperatures, hypoxia, changes in salinity, and wave action, all of which can mediate species interactions in predictable manners (Sanford 1999, Wetthey 2002, Van de Koppel et al. 2006). Here we focus on body temperature (during both low and high tide) as it can drive dynamics at multiple levels (molecular, organismal, assemblage) of biological organization (Hochachka and Somero 2002). Importantly, intertidal factors, such as temperature, are inherently variable in time and space (Broitman et al. 2009, Denny et al. 2011) as a result of both the influence of habitat complexity as well as of how organisms exchange heat with their local environments (Monaco and Helmuth 2011, Potter et al. 2013, Sunday et al. 2014). Furthermore, because the thermal niche often varies between interacting species, exposures to similar temperatures may affect them differently (Somero 2002). In order to correctly determine whether consumers or prey are more affected by environmental stress, it is essential to account for this variability, which can be especially problematic for mobile species (Huey 1991). Caging experiments or other methods where movement is restricted provide a means for manipulating and testing the effect of environmental stress gradients (Menge et al. 2002, Petes et al. 2008), but because these methods impair an organism's natural behavior, they may potentially yield misleading results.

Here we describe an alternative approach to circumvent this problem, which relies on field observations of individuals' microhabitat use, biomimetic temperature logger records (Fitzhenry et al. 2004, Szathmary et al. 2009, Seabra et al. 2011), and the thermal performance curve (TPC) framework (Huey and Stevenson 1979, Huey and Kingsolver 1989, Woodin et al. 2013). Biomimetic temperature loggers are commercial sensors that have been modified to estimate the morphological and thermal properties of an organism, and therefore capture its body temperature with relatively high accuracy. Thermal performance curves describe the dependence of organisms' vital rates (e.g., metabolism, feeding, growth, reproduction) on temperature (Angilletta 2009). By quantifying thermal performance indirectly using TPCs and *in situ* continuous measurements of body temperature, one can avoid influencing an organism's behavior due to experimental manipulations. Combined with measurements of behavior as a function of environmental conditions, one can then estimate realized thermal performance (Huey et al. 2009).

Using this approach for multiple species permits a comparison of thermal performance among interacting species (Freitas et al. 2007, Dell et al. 2011). These comparisons provide an estimate of relative performance under current climate scenarios (Somero 2010), and thus an explicit test of when CSMs and PSMs are most likely to occur (Monaco and Helmuth 2011). Thermal responses

have been quantified for numerous species of microbes, plants, and animals covering all major ecosystems; and the relative performance of these interacting species may depend on taxa, trophic levels, functional traits, and overall context (Dell et al. 2011). For instance, foraging strategy of consumers (e.g., active-capture, sit-and-wait, grazing) will determine which functional traits (e.g., relative velocities of consumers and resources) need to be addressed for specific pairs of species (Dell et al. 2013), which can ultimately define the nature of an ESM as CSM or PSM. The attack speed of predators, for example, can have different thermal sensitivities than the escape speed of their prey (Grigaltchik et al. 2012). This application of TPCs is far less common for intertidal species pairs (Monaco and Helmuth 2011).

The predictive power of this framework for informing community-level processes can be further improved when applied to keystone and structuring species, whose dynamics may disproportionately influence community function, stability, and biodiversity (Crain and Bertness 2006, Smith et al. 2006). To examine potential direct and indirect effects of temperature, we focused on a major predator–prey interaction in rocky shores of the Pacific coast of North America, the predatory sea star *Pisaster ochraceus* (hereafter, *Pisaster*) and its main prey, the mussel *Mytilus californianus* (hereafter, *Mytilus*). By foraging on *Mytilus*, a dominant competitor for space, *Pisaster* facilitates the presence of other large space-occupying invertebrates and algae, thus fulfilling a keystone ecological role (Paine 1966, 1974). Conversely, as a structuring species, the presence of *Mytilus* beds has significant follow-on effects on diversity of smaller organisms living within the bed (Smith et al. 2006). Recent work has demonstrated that temperatures during high and low tides (and their interaction) may drive the strength of interaction between *Pisaster* and *Mytilus* by altering rates of foraging (Sanford 1999, Pincebourde et al. 2008, 2012). Specifically, while increases in water temperature are associated with increased foraging rates (Sanford 1999) increases in aerial body temperature (which are generally considerably higher than those in water) can reduce feeding rate up to 40% (Pincebourde et al. 2008). These results point to the importance of considering the physiological basis of these temperature effects.

By combining data on metabolic rates during submergence/exposure periods, Fly et al. (2012) quantified energetic costs of occupying different shore levels for *Pisaster*, and found no marked differences between being in the mid-high intertidal zone vs. lower on the shore. Following the ESM framework, Petes et al. (2008) experimentally tested which species was more affected by the environment in both the low and high zones, concluding that the system behaved as a CSM. However, their method of caging individuals might have influenced their results, particularly because they constrained the ability of *Pisaster* to move among microhabitats thereby potentially ameliorating stress (Huey 1991).

Here we examined the physiological performance of both species (*Pisaster* and *Mytilus*) revisiting the question of which is more likely to be negatively impacted by environmental stress under any given suite of environmental conditions. First, to explore the role of physiology, we combined information of empirically derived TPCs with field observations of realized body temperatures. Here we use aerobic metabolic rate as a measure of relative physiological performance, as it relates to an organism's ability to utilize and process energy (Schulte et al. 2011, Verberk et al. 2015). Second, to evaluate the role of movement behavior in *Pisaster* on its mean thermal performance, we included observations of microhabitat use as a function of environmental conditions. And third, we complemented this with empirical indicators of overall physiological condition (body mass index, BMI) and heat stress (heat-shock protein 70 kDa production, Hsp70), hypothesizing that greater thermal stress is directly related to Hsp70 and inversely related to BMI. To test whether results were generalizable across sites, these analyses were conducted at two locations (Bodega Marine Reserve, California, and Strawberry Hill, Oregon) ~760 km apart, with contrasting thermal environments. Because the times of the lowest low tides are closer to noon during summer at the latter location (Helmuth et al. 2002, Place et al. 2008), we predicted greater thermal stress there, analogous to what could be expected on high-shore intertidal levels (Stillman and Somero 2000).

METHODS

Study sites

We conducted field surveys, collected tissue samples, and collected animals for lab experiments at two sites: Strawberry Hill (44°14'59.4" N, 124°06'54.7" W, Oregon, USA), and Bodega Marine Reserve (38°19'07.7" N, 123°04'27" W, California, USA). We chose these sites because (1) *Pisaster* and *Mytilus* were highly abundant and interacting widely at both sites; (2) environmental conditions were expected to be dissimilar, given the time of the lowest low tides being closer to midday at Strawberry Hill (Helmuth et al. 2002); and (3) the physical habitat is topographically complex at both sites, offering alternative microhabitats for *Pisaster* to seek refuge (crevices, tide pools, kelps, open spaces). At the time of the study (2012), wasting disease had not yet impacted sea star populations (Hewson et al. 2014, Eisenlord et al. 2016).

Field body temperature measurements

We used biomimetic temperature loggers customized to resemble the thermal properties of average-sized *Pisaster* (~9 cm arm length; Pincebourde et al. 2008, Szathmary et al. 2009) and *Mytilus* (~8 cm shell length; Fitzhenry et al. 2004, Broitman et al. 2009). While *Pisaster* preferentially forages on mussels <8 cm shell length (Paine 1976), our biomimetic design for *Mytilus* could not be made any

smaller due to size constraints set by the commercial temperature logger used (TidBit; Onset Computer Corporation, Bourne, MA, USA) (Fitzhenry et al. 2004). Nevertheless, the effect of mussel size (4–8 cm shell length range) on biomimetic temperature has been previously shown to be small (Helmuth 1998). Sea star and mussel biomimetics, also known as “robo-sea stars” and “robo-mussels,” respectively, recorded potential body temperatures between 22 June and 10 August 2012, once every 30 min, at a resolution of 0.2°C, using TidBit data loggers.

Because *Pisaster* can move throughout the intertidal zone, we deployed robo-sea stars in microhabitats where sea stars were common: directly exposed to the elements (solar radiation, wind) in the mid-intertidal zone, sheltered from solar radiation and wind in crevices, or submerged in tide pools. Robo-mussels, in turn, were deployed only in the mid-intertidal, where *Mytilus* is stationary and their interaction with *Pisaster* is strongest. All biomimetics were deployed ~1.2 m above mean lower low water (MLLW).

Surveys of Pisaster microhabitat use

To determine *Pisaster* microhabitat use during the period of the study, we conducted surveys on five different low spring-tide periods over the summer of 2012 at Bodega (27 May, 7 June, 25 June, 23 July, and 2 August) and Strawberry Hill (26 May, 7 June, 23 June, 22 July, and 5 August). The microhabitat of every individual sea star encountered within five, 2 m wide, belt transects oriented perpendicularly to the coastline from the highest sea star to the water line was recorded. We categorized each individual based on microhabitat use as sun-exposed (unprotected from direct solar radiation and wind), sheltered from solar radiation in crevices, or submerged in tide pools. Our surveys thus deliberately ignored *Pisaster* behavior during neap tide periods (Feder 1956), but since this species remains largely submerged during this time we assumed body temperatures matched seawater temperatures. While foraging activity may increase during neap tide periods (Hayford et al. 2015), we were ultimately interested in estimating overall thermal performance, which theoretically integrates all physiological processes, including feeding (Dell et al. 2011, Schulte et al. 2011). By using records of both aerial and submerged body temperature collected over the entire lunar cycle combined with TPCs, we were able to estimate relative performance during both neap and spring tide periods (see *Theoretical quantification of physiological performance*).

Empirical indicators of physiological performance

Body mass indices.—We calculated body mass indices for both *Pisaster* (10 animals/site) and *Mytilus* (40 animals/site). Individuals of average size were collected at Bodega on 19 July 2012 (*Pisaster* arm length [mean ± SE], 11.21 ± 1.98 cm; *Mytilus* shell length, 4.81 ± 0.06 cm), and Strawberry Hill on 22 July

2012 (*Pisaster* arm length, 10.12 ± 1.19 cm; *Mytilus* shell length, 5.13 ± 0.07 cm), and transported fresh to Bodega Marine Laboratory (BML, UC-Davis, Bodega Bay, CA, USA) for analyses. We dissected sea stars, separating gonads and pyloric caecum from the body walls. We determined the dry mass of gonads (GDM), pyloric caeca (PDM), and body walls (BwDM) by drying at 80°C for 48 h, and weighing them to the nearest 0.001 g. *Pisaster* BMI was calculated as $(GDM + PDM) \times (GDM + PDM + BwDM)^{-1}$. We dissected mussels by separating all soft tissue from the shell. To determine the dry mass of tissue (TDM) and shell (ShDM), we dried them at 80°C for 24 h, and weighed to the nearest 0.001 g. We calculated *Mytilus* BMI as $(TDM) \times (TDM + ShDM)^{-1}$.

Heat shock protein expression.—We measured heat shock protein 70 kDa (Hsp70) expression from sea star (5 animals·site⁻¹·sampling⁻¹) and mussel (6 or 7 animals·site⁻¹·sampling⁻¹) tissue samples collected during the same five spring-tide periods when microhabitat use surveys were conducted at Bodega and Strawberry Hill. All samples were collected during spring tides. We chose individuals found on the lower edge of the mussel bed (mid-intertidal zone), where these species interact most frequently. Tissue samples were removed *in situ* (tube-feet for *Pisaster*, and gills for *Mytilus*), and immediately frozen using dry ice. Within 24 h of sampling, tissues were stored at BML in -80°C freezers. We shipped the samples on dry ice over night to the University of South Carolina, where they were stored again at -80°C for subsequent immunochemical detection of Hsp70.

We quantified Hsp70 expression using dot blot analysis. Previously, we had optimized the concentrations of protocol constituents using western blot analysis, as modified from Hofmann and Somero (1995) and Helmuth and Hofmann (2001). Although we were unable to discriminate between constitutive and inducible Hsp70 isoforms on our gels, preliminary assessments of *Pisaster* heat shock response after short-term (i.e., days) high temperature treatments revealed increases in Hsp70 expression, which can be considered as changes in the inducible isoform given the temporal resolution of the experiments (M. Kinsey and S. Place, *personal communication*). Therefore, we regarded our measurements as total Hsp70 (i.e., constitutive + inducible) expression. Because we were interested in dynamics occurring over weeks and months, not distinguishing between the two isoforms did not impair our ability to examine *Pisaster* and *Mytilus* heat shock response.

We homogenized samples (~0.05 g) in 0.5 mL of homogenizing buffer (50 mmol/L Tris-HCl pH 6.8, 4% [w/v] SDS, complete protease inhibitor cocktail). Homogenates were incubated at 100°C for 5 min, centrifuged at $12000 \times g$ for 15 min, and the supernatant stored at -20°C. To determine total protein concentration of aliquots reserved from the sample extracts, we used a Bradford protein assay (Pierce Coomassie Plus, ThermoFisher Scientific, Waltham, MA, USA).

We loaded 10 μg of extracted proteins onto a hydrated 0.2- μm nitrocellulose membrane placed flat in a 96-well dot blot apparatus (Bio-Rad Laboratories, Hercules, CA, USA). Samples were allowed to migrate by gravity for 30 min. Blotted membranes were washed in phosphate-buffered saline (PBS; 8.1 mmol/L Na_2HPO_4 , 2.7 mmol/L KCl, 137 mmol/L NaCl, 1.5 mmol/L KH_2PO_4 , pH 7.4) for 10 min, blocked (blocking solution; 5% non-fat dry milk in PBS-Tween20 0.1%) for 1 h, and washed in PBS-Tween20 0.1% for 5 min three times. We incubated the blots in a 1:2000 dilution of primary antibody solution (ENZO anti-Hsp70 pAb-ADI-SPA-757, 80% blocking solution, 20% fetal bovine serum, 0.02% thimerosal, 1 mmol/L PMSF) for 1.5 h, washed in PBS-Tween20 0.1% for 5 min three times, and then incubated in a 1:6000 dilution of secondary antibody solution (Santa Cruz goat anti-rabbit IgG-HRP-SC2004, blocking solution) for 1 h. Blots were washed once in PBS-Tween20 0.3% for 5 min, twice in PBS-Tween20 0.1% for 5 min, and once in PBS for 5 min. Next, we incubated them in an enhanced chemiluminescence reagent (ECL; ThermoFisher Scientific SuperSignal, Waltham, MA, USA) for 5 min, exposed films for 40 min, and digitized them using an imaging system (Fotodyne, Hartland, WI, USA). Dot intensity was determined using the software ImageJ, (US National Institute of Health, Bethesda, MD, USA). We calculated relative values of Hsp70 for the samples based on readings obtained from positive controls (purified recombinant Hsp70, ENZO ADI-SPP-758), which were loaded along with the tissue samples in every dot blot.

Theoretical quantification of physiological performance

Estimating thermal performance curves.—We first parameterized aquatic thermal performance curves for both *Pisaster* and *Mytilus*, from Bodega and Strawberry Hill, using empirical metabolic rate data. We estimated aerial thermal sensitivity curves, using data on physiological responses to temperature under exposed conditions obtained from the literature coupled with key parameters from the aquatic respiration measurements (described in further detail under *Aquatic thermal performance*). We fitted all four curves based on formulations by Stevenson et al. (1985):

$$P(T_b) = S \times (1 / (1 + k_1 \times e^{-k_2 \times (T_b - CT_{\min})})) \times (1 - e^{k_3 \times (T_b - CT_{\max})}), \quad (1)$$

which explicitly discriminate between a lower and a higher region of the response curve, where physiological rates increase and decrease, respectively. Here, P is a physiological rate at body temperature T_b , S is a scaling parameter, k_1 , k_2 , and k_3 are constants, and $CT_{\min}$ and $CT_{\max}$ are the lower and upper limits to performance. The scaling parameters and constants in Eq. 1 were empirically estimated (see *Aquatic thermal performance* and *Aerial thermal performance*). To specifically compare thermal sensitivity at temperatures where enzymes are active

between populations, we also estimated the Arrhenius temperature, T_A (analogous to Q_{10} ; Freitas et al. 2007).

Aquatic thermal performance.—Sea stars and mussels were collected at Strawberry Hill (24 June 2012), stored in coolers packed with kelp and icepacks on the bottom to maintain cool and humid conditions, and transported by ground to BML (~11 h trip). We then acclimated *Pisaster* and *Mytilus* in separate tanks with running seawater at ambient temperature (~12°C) for 5 d. This relatively short acclimation was not intended to remove any possible endogenous rhythms associated to the two-week lunar cycle (Hayford et al. 2015), but to ensure individuals were healthy before exposure to treatment conditions. Food was supplied *ad libitum* for both sea stars (mussels provided in excess) and mussels (IAP Algae Paste, (Innovative Aquaculture Products, Skerry Bay, BC, Canada), Spat Formula; 10% dilution frozen into ice cubes and placed in tanks while stopping the water flow for 2 h, twice a day; Schneider et al. 2010). The same protocol was followed for individuals collected at Bodega (7 July 2012), though instead of transporting the animals, we kept them in coolers with kelp and icepacks for the same time as those collected at Strawberry Hill.

The thermal performance of both *Pisaster* and *Mytilus* was determined from metabolic rate measurements taken at six temperatures: 10°, 13°, 18°, 21°, 24°, and 27°C. Following the acclimation period, we placed two individual *Pisaster* and two *Mytilus* in 60-L aquaria (three per treatment) filled with 1- μm filtered seawater at ambient temperature (~12°C). Treatment seawater temperatures were adjusted by keeping the aquaria in climate-controlled rooms available at BML. The two highest temperatures were reached using 100-W aquarium heaters (Marineland Visi-Therm, USA Blacksburg, VA). Water temperatures were changed at a rate of ~1°C/h. We kept the individuals at their treatment temperatures for 4 d, after which we measured oxygen consumption rates. To ensure water quality, tanks were fitted with air stones and submersible pumps. Water chemistry (salinity, pH, ammonia, nitrite, and nitrate) was monitored every other day using a saltwater test kit (API, USA Philadelphia, PA), and partial water changes were performed when necessary (every 1–2 d).

To measure oxygen consumption, we placed individual sea stars and mussels in watertight chambers (2.88 and 0.7 L, respectively) filled with aerated, 1- μm filtered seawater, at its corresponding treatment temperature. A magnetic stir-bar kept the water circulating during measurements. Animals had limited room to move in the chambers. Over the top of each chamber, we fitted Clark-type electrodes (HANNA-9146, USA, Hanna Instruments, Bedfordshire, UK), and measured dissolved oxygen concentration (mg/L) at 10 and 40 min after sealing the chamber. Trials were discontinued if oxygen levels dropped below 70% of initial readings so that metabolic rate would not be affected by depletion of dissolved oxygen. To control for background variability in

oxygen content, we conducted measurements in two animal-free chambers at each of the treatment temperatures. We standardized the change in oxygen concentration by the animal's dry mass, and expressed as standard metabolic rate (SMR, $\mu\text{mol O}_2 \cdot \text{h}^{-1} \cdot \text{g DM}^{-1}$). The experimental design yielded three replicates per temperature per site for *Pisaster* and *Mytilus*. All animals maintained at the warmest treatment temperature, 27°C, died within two days of beginning the thermal conditioning period, so a value of zero was assigned to them when fitting thermal performance curves.

To estimate aquatic TPC parameters, we normalized the oxygen consumption data for each treatment by the highest value. To estimate the parameters S , k_1 , k_2 , k_3 , $\text{CT}_{\min}$, and $\text{CT}_{\max}$ we used the Levenberg-Marquardt nonlinear least squares optimization method (R package `minpack.lm`) (Moré 1978). We estimated the parameter T_A from the slope of a linear model between $\ln(\text{SMR})$ and the inverse of treatment temperature in Kelvin, for the range of temperatures where SMR increased exponentially (Freitas et al. 2007).

Aerial thermal performance.—Data to fit aerial thermal performance curves for each species and site were obtained from the literature. Although information was not available for the whole thermal range, key parameters that constrain the curves (e.g., lethal temperatures) were obtained from literature, which, combined with the aquatic thermal performance parameters from each site, allowed fitting site-specific aerial thermal performance curves for each species. For *Pisaster*, we obtained data of aerial physiological rates relative to measurements taken in water at the same temperatures from Fly et al. (2012) and critical temperatures from (Monaco et al. 2015) and Pincebourde et al. (2008). For *Mytilus*, we obtained data on aerial physiological rates relative to measurements taken in water at the same temperatures from Bayne et al. (1976). Critical temperatures were taken from Denny et al. (2011) and Mislán et al. (2014). Note that Mislán et al. (2014) also found that upper critical temperature of *Mytilus* collected at Bodega did not differ from that of individuals collected at Boiler Bay, Oregon, a site located relatively close to Strawberry Hill (~70 km), suggesting no difference in upper thermal limits between the latter and Bodega.

The manual fitting protocol involved an extensive grid search, by which the combination of parameter values for S , k_1 , k_2 , k_3 , and $\text{CT}_{\min}$ that minimized the difference between observed and predicted data according to root mean square errors (RMSEs). $\text{CT}_{\max}$ was fixed based on the literature (Bayne et al. 1976, Pincebourde et al. 2008, Denny et al. 2011, Fly et al. 2012, Mislán et al. 2014).

Calculating mean thermal performance.—We calculated *Pisaster* and *Mytilus* relative thermal performance at Bodega and Strawberry Hill for the time period when our biomimetic sensors were deployed. The thermal performance models estimated for each species were run

using the temperature records from each biomimetic sensor and tide height data (to inform when loggers were submerged/emerged) as inputs, thus generating relative performance time-series for the prey and each of the microhabitats where the predator is found. We downloaded the tide height data from NOAA's CO-OPS (station IDs 9435380 and 9415020 for Strawberry Hill and Bodega, respectively).

Accounting for *Pisaster* behavior.—We evaluated the role of movement on *Pisaster* thermal performance by recreating three scenarios: mobile, static, and optimal predator. The static scenario was computed using the thermal performance curves and the mid-intertidal robo-sea stars. For the mobile scenario, i.e., to account for *Pisaster* movements throughout the intertidal zone during the period when temperature measurements were taken, we ran 500 simulations in which hypothetical individuals were allowed to instantaneously choose between microhabitats (exposed mid-intertidal, crevice, tide pool) every 12 h. Their selection was limited by the probability of occupying a specific microhabitat, as informed by our surveys conducted during the same period. Note that although microhabitat use surveys were only conducted during spring tide periods, *Pisaster* remains largely submerged during neap tide periods (Feder 1956), so information of seawater temperatures are representative of body temperatures. Movement between microhabitats was computed instantaneously, so any potential change in thermal performance during the sea star displacement period was ignored in our calculations. For the optimal scenario we hypothetically allowed *Pisaster* to instantaneously move to the microhabitat reporting the highest performance for each time point.

Statistical analyses

We ran all calculations and statistical analyses using R 3.0.1 (R Core Team 2013). From the field body temperatures recorded via biomimetic loggers we determined the daily maximum values and compared between site and species using a two-way ANOVA. In addition to recording the extreme high value recorded during the record, to explore temporal variability in temperature records for predator and prey, we calculated average daily range (ADR). We compared degree of variability between sites/species using F ratio tests (Quinn and Keough 2002). We also calculated the thermal safety margin ($\text{TSM} = \text{CT}_{\max} - \text{Extreme}$) for predator and prey to examine risk of temperature-related mortality (Sunday et al. 2014).

The thermal sensitivity parameter T_A was compared between sites and species using one-tailed z -score tests. To compare relative thermal performance data between species/scenarios at each site, we calculated RMSEs and to test for correlation between them, we computed Kendall's W coefficient of concordances. To compare temporal variability between species/scenarios and sites, we conducted F ratio tests (Quinn and Keough 2002).

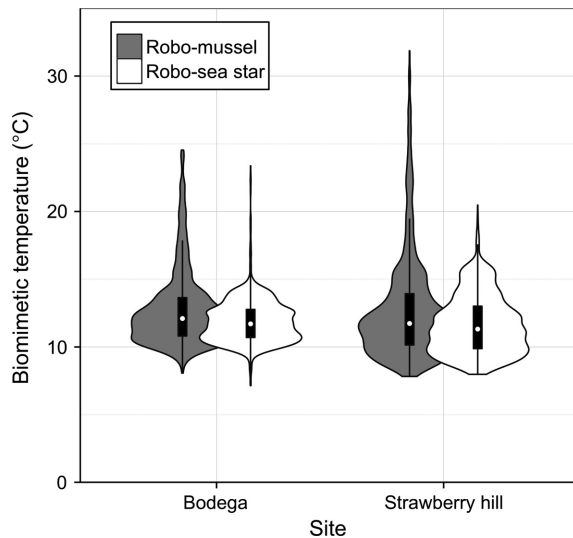


FIG. 1. *Pisaster ochraceus* and *Mytilus californianus* biomimetic temperatures recorded in the mid intertidal zone at Bodega and Strawberry Hill during summer months of 2012 (22 June through 10 August). Data were collected every 30 min. The 'violin' shapes represent the distribution densities of the data. The white circles are the medians. The boxes are 25th and 75th percentiles of the distribution. The lines extending vertically are 1.5 times the interquartile range.

We compared BMI between sites (categorical) for each species using Welch's two sample *t* tests because data were heteroscedastic (Levene test, $P < 0.01$). We analyzed Hsp70 data separately for each species using two-way ANOVAs. For the lab experiment data, temperature (continuous) and site were considered as fixed factors, whereas for the field collected data, date (categorical) and site were the fixed factors. When assumptions of normality and homoscedasticity were not satisfied for Hsp70 data (even after log-transformations), we ran two ANOVAs, one using raw data and another using rank-transformed data. If results were qualitatively the same between the tests, we report results from the former; otherwise we provide results from the latter.

RESULTS

Biomimetic temperature records

When comparing daily maximum body temperatures recorded using biomimetic loggers, we found significant effects of both species (two-way ANOVA; $F_{1,196} = 50.78$, $P < 0.001$) and sites (two-way ANOVA; $F_{1,196} = 8.42$, $P = 0.004$). Mussels experienced higher extreme temperatures than sea stars at both sites, and Strawberry Hill appeared, on average, warmer than Bodega for both species; while the highest temperature recorded for mussels was at Strawberry Hill, the extreme for sea stars was higher at Bodega Bay (Fig. 1, Table 1). Note that daily maximum body temperatures do not discriminate between aquatic and aerial conditions. Thus, while during spring tide

periods these likely correspond with aerial exposure, during neap tide periods they probably represent the maximum daily water temperature. Temporal variability in biomimetic temperature records was greater for *Mytilus* than *Pisaster* (*F* ratio test; $F = 3.19$, $P < 0.001$, *df* (4613, 4613)), and for both species variability (average daily range) was greater at Strawberry Hill than Bodega (*F* ratio test; $F = 0.52$, $P < 0.001$; Fig. 1, Table 1). Thermal safety margins revealed that, at Bodega, *Pisaster* had a slightly smaller (1.8°C) TSM than *Mytilus*, while at Strawberry Hill, *Mytilus* was at a substantially higher (8.4°C) risk (Table 1).

Theoretical indicators of physiological performance

Thermal performance curves.—*Pisaster* and *Mytilus* mean metabolic rate increased with water temperature up to a maximum point between 20° and 25°C, varying with species and site (Fig. 2). All animals died in the 27°C treatment. These data allowed fitting aquatic thermal performance curves (Eq. 1, Table 2) to each species/site using Levenberg-Marquadt nonlinear optimization models (Fig. 3).

Using the temperature range where SMR increased, we fitted linear regression models between $\ln(\text{metabolic rate})$ and temperature in Kelvin, and determined the slopes, which represent the parameter Arrhenius temperature, T_A (Freitas et al. 2007), for each site and species (Table 2). The parameter T_A was higher at Bodega than Strawberry Hill for *Pisaster* ($z = 3.41$, $P < 0.001$), but did not differ between sites for *Mytilus* ($z = -0.82$, $P = 0.21$). At Bodega, *Pisaster* showed a higher thermal sensitivity (i.e., higher T_A) than *Mytilus* ($z = 1.76$, $P = 0.04$). At Strawberry Hill, in contrast, *Mytilus* showed a higher thermal sensitivity ($z = -2.16$, $P = 0.02$).

A parameter grid search allowed fitting aerial thermal performance curves (Fig. 3, Table 2) using the aquatic thermal performance parameters as baselines and data on aerial metabolic rates obtained from the literature.

TABLE 1. *Pisaster* (predator) and *Mytilus* (prey) body temperature variability metrics, as recorded by biomimetic loggers deployed at Bodega and Strawberry Hill: average daily maximum (ADM; mean $\pm$ SE), average daily range (ADR), extreme high, and thermal safety margin (TSM = $CT_{\text{max}} - \text{Extreme}$).

Animal	ADM	ADR	Extreme	TSM
Bodega				
Prey	19.17 $\pm$ 0.39	8.73	24.56	13.44
Predator	14.30 $\pm$ 0.40	3.72	23.39	11.61
Optimal predator	14.67 $\pm$ 0.40	3.61	23.39	11.61
Strawberry Hill				
Prey	21.02 $\pm$ 0.69	11.19	31.89	6.11
Predator	15.05 $\pm$ 0.30	5.05	20.52	14.48
Optimal predator	15.58 $\pm$ 0.53	5.23	25.35	9.65

Notes: Temperature records do not discriminate between aquatic and aerial conditions. Metrics are provided for stationary mussels and sea stars, as well as for the optimal movement scenario calculated for sea stars.

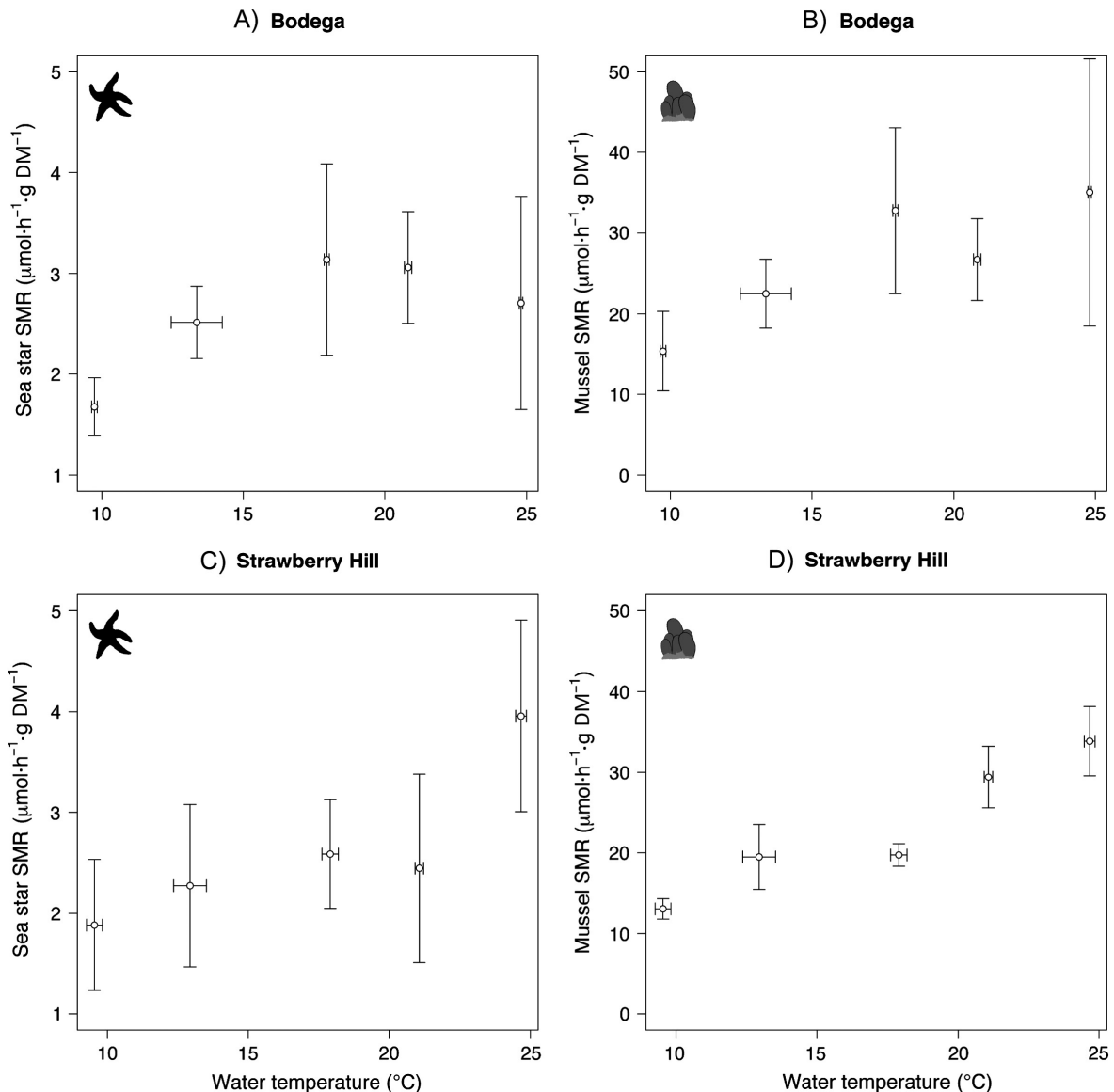


FIG. 2. Standard metabolic rate (SMR) of the sea star *Pisaster ochraceus* and mussel *Mytilus californianus* collected from Bodega and Strawberry Hill. No data are provided for the 27°C treatment because all animals died during the first two treatment days. Values are means $\pm$ 1 SD. DM, dry mass.

Relative thermal performance.—At Bodega, we observed that relative (scaled) performance (mean $\pm$ SE) was lower for *Mytilus* (0.55 ± 0.002) than any of the *Pisaster* scenarios evaluated: static (0.65 ± 0.003), mobile (0.67 ± 0.0002), and optimal (0.71 ± 0.002), with RMSEs = 0.16, 0.19 ± 0.0001 , and 0.19, respectively (Fig. 4A, Table 3). No clear differences were detected between the three *Pisaster* scenarios, although the optimal was slightly higher than the static and mobile scenarios (RMSE = 0.13 and 0.10 ± 0.0004 , respectively). Concordances between species/scenarios were significant in all cases, revealing strong association between them ($P < 0.01$ in all cases; Table 3).

At Strawberry Hill, the patterns were comparable to Bodega. Relative performance was lower for *Mytilus* (0.41 ± 0.002) than for any of the *Pisaster* scenarios: static (0.44 ± 0.002), mobile (0.43 ± 0.00008), and optimal (0.46 ± 0.002), with RMSEs = 0.09, 0.10 ± 0.00008 , and 0.09, respectively (Fig. 4B, Table 3). The *Pisaster* scenarios showed no marked differences, except that the optimal was higher than static and mobile (RMSEs = 0.07 and 0.09 ± 0.00008 , respectively). Also at Strawberry Hill, we observed significant concordance between species/scenarios ($P < 0.01$ in all cases; Table 3).

Pisaster and *Mytilus* mean thermal performance was higher at Bodega for every scenario (Fig. 4). Temporal

TABLE 2. Thermal performance curves parameter values estimated for *Pisaster* and *Mytilus* individuals collected at Bodega and Strawberry.

Parameter	Bodega		Strawberry Hill	
	Aquatic	Aerial	Aquatic	Aerial
<i>Pisaster ochraceus</i>				
S	0.88 ± 0.06	0.60 (0.01)	0.66 ± 0.32	0.60 (0.04)
k_1	$2.79 \pm 6.5 \times 10^7$	3.20 (0.01)	$5.66 \pm 1.06 \times 10^8$	5.20 (0.04)
k_2	0.39 ± 0.17	0.30 (0.01)	0.17 ± 0.13	0.10 (0.04)
k_3	0.88 ± 0.32	0.60 (0.01)	$2.41 \pm 2.79 \times 10^{23}$	0.30 (0.04)
$CT_{min}, ^\circ C$	$6.92 \pm 6.4 \times 10^7$	6.91 (0.01)	$1.42 \pm 1.12 \times 10^8$	1.42 (0.04)
$CT_{max}, ^\circ C$	27 ± 0.08	35 (0.01)	27 ± 0.005	35 (0.04)
T_A, K	6221 ± 778	NaN	3182 ± 435	NaN
<i>Mytilus californianus</i>				
S	0.76 ± 0.11	0.6 (0.03)	1.02 ± 0.33	0.6 (0.04)
k_1	$7.21 \pm 1.36 \times 10^8$	6 (0.03)	$6.92 \pm 5.95 \times 10^7$	6 (0.04)
k_2	0.31 ± 0.15	0.3 (0.03)	0.17 ± 0.07	0.2 (0.04)
k_3	$12.44 \pm 4.12 \times 10^5$	0.2 (0.03)	$10.45 \pm 2.15 \times 10^2$	1 (0.04)
$CT_{min}, ^\circ C$	$4.14 \pm 6.47 \times 10^7$	4.14 (0.03)	$3.02 \pm 6.03 \times 10^7$	3.02 (0.04)
$CT_{max}, ^\circ C$	27 ± 9.85	38 (0.03)	27 ± 9.85	38 (0.04)
T_A, K	4187 ± 853	NaN	5140 ± 794	NaN

Notes: Aquatic parameters (mean $\pm$ SE) were estimated using Levenberg-Marquadt optimization. Aerial parameters (mean with total root mean square error [RMSE] in parentheses) were estimated through a grid search. NaN, parameter was not estimated for aerial conditions. S is a scaling parameter, k_1 , k_2 , and k_3 are constants, CT_{min} and CT_{max} are the lower and upper limits to performance, and T_A is the Arrhenius temperature. See *Materials and Methods* for a description of estimation procedures.

variability in thermal performance was also greater at Bodega for both species and for every scenario: *Mytilus* (F ratio test; $F = 1.20$, $P < 0.001$), *Pisaster* static (F ratio test; $F = 3.74$, $P < 0.001$), *Pisaster* mobile (F ratio test; $F = 2.77$, $P < 0.001$), and *Pisaster* optimal (F ratio test; $F = 1.50$, $P < 0.001$).

Note that allowing *Pisaster* to move in the optimal scenario did not significantly change the amount of time spent submerged (Bodega = 84.97%, Strawberry Hill = 86.06%), relative to the static scenario (Bodega = 79.84%, Strawberry Hill = 80.92%).

Empirical indicators of physiological performance

Body mass indices.—*Pisaster* BMI showed no differences between sites (Fig. 5A; Welch's t test, $t_{17,25} = -0.76$, $P > 0.05$). By contrast, *Mytilus* BMI was significantly higher at Strawberry Hill than Bodega (Fig. 5B; Welch's t test, $t_{52,32} = -11.39$, $P < 0.01$).

Heat shock protein (Hsp70) expression.—Lab experiments showed that *Pisaster* expression of Hsp70 was not affected by temperature (two-way ANOVA; $F_{4,50} = 0.11$, $P > 0.05$), site ($F_{1,50} = 0.01$, $P > 0.05$), and their interaction ($F_{4,50} = 0.13$, $P > 0.05$). Likewise, for *Mytilus* we did not see a detectable change in expression of Hsp70 with temperature (two-way ANOVA, $F_{4,48} = 0.48$, $P > 0.05$), site ($F_{1,48} = 0.002$, $P > 0.05$), and their interaction ($F_{4,48} = 2.42$, $P > 0.05$) (Fig. 6A, B).

Our field tissue samples revealed that *Pisaster* Hsp70 production remained constant across survey dates (two-way ANOVA; $F_{4,41} = 1.18$, $P > 0.05$) and sites ($F_{1,41} = 1.76$, $P > 0.05$), and their interaction was nonsignificant ($F_{4,41} = 0.81$, $P > 0.05$; Fig. 6C, D). In turn, *Mytilus* Hsp70 production in the field varied with survey

date (two-way ANOVA; $F_{4,54} = 5.98$, $P < 0.01$), but did not change between sites ($F_{1,54} = 0.42$, $P > 0.05$), and no interaction between them was detected ($F_{4,54} = 2.45$, $P > 0.05$). When comparing level of temporal variability between sites, levels for *Mytilus* were significantly greater at Strawberry Hill ($SD = 0.41$) than Bodega ($SD = 0.15$; F ratio test; $F = 0.14$, $P < 0.001$; Fig. 6C, D).

DISCUSSION

By combining information of organisms' thermal performance, their potential body temperatures experienced in the field, and estimates of thermoregulatory behavior, this study provides a unique perspective of the thermal physiology of two key species from the Pacific coast of North America, the predator *Pisaster ochraceus* and its main prey *Mytilus californianus*. Furthermore, embedded within an environmental stress model framework, this approach stands as a powerful tool to uncover and predict mechanisms driving ecological dynamics associated with this and other predator–prey systems. While similar approaches have long been applied in terrestrial systems for organisms such as reptiles (Huey et al. 1989, Kearney et al. 2008) the approach has not been used extensively in marine systems nor has it previously been used in a comparative framework to explore ESMs.

Body temperatures

By measuring temperatures using thermally matched biomimetic sensors, we captured the real conditions that *Pisaster* and *Mytilus* would have potentially experienced in the field in different microhabitats (Fitzhenry et al. 2004, Szathmary et al. 2009). These records indicated that extreme high body temperatures as well as more

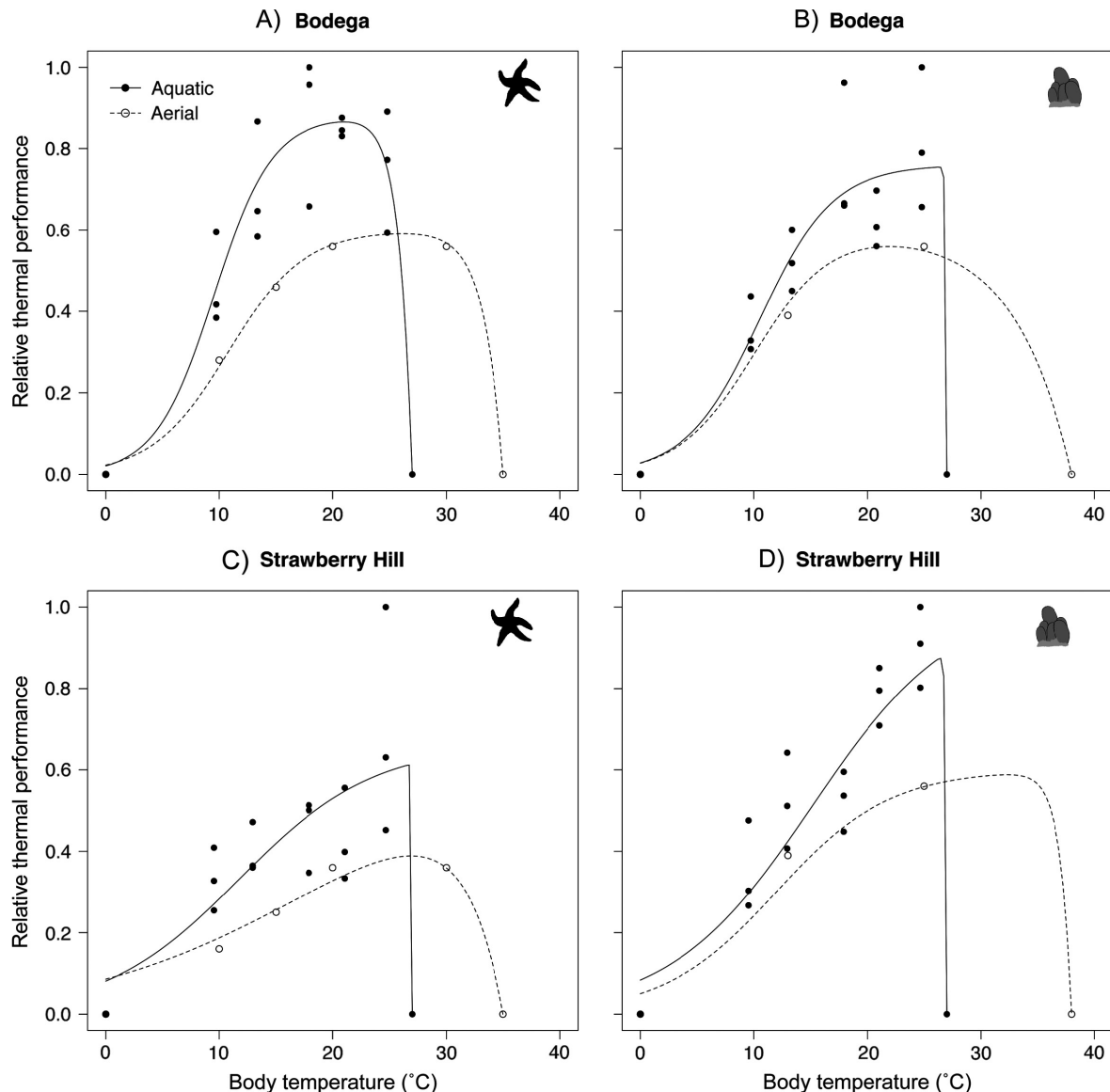


FIG. 3. Aquatic (solid) and aerial (dashed) thermal performance curves described for *Pisaster ochraceus* and *Mytilus californianus* from Bodega (Panels A and B) and Strawberry Hill (Panels C and D). We estimated the aquatic curves' parameters using empirical SMR measurements (Fig. 2), and the aerial curves' parameters using information from the literature.

chronic daily maxima differed between species and sites (Fig. 1). On-average higher daily temperature extremes at Strawberry Hill, observed for both species, are likely due to the timing of low tide being closer to noon during the summer months (Helmuth et al. 2002, Place et al. 2008). Notably, rare, extreme temperatures were greater for mussels at Strawberry Hill but were greater for sea stars at Bodega. These results are in line with previous studies showing that while sea stars generally maintain lower body temperatures than mussels due to the former species' ability to cool through evaporation, the relative difference in temperature between species is site-dependent (Broitman et al. 2009). Temporal variability

in body temperature also differed between species and sites, which, mediated by differential thermal sensitivities, would determine differences in physiological performance.

Both robo-sea stars and robo-mussels mimicked animals slightly larger than the average size *Pisaster* and *Mytilus* (Fitzhenry et al. 2004, Pincebourde et al. 2008, Broitman et al. 2009, Szathmary et al. 2009). Capturing body temperatures across size classes would potentially provide more accurate and complete results, although prior modeling efforts (Helmuth 1998) have suggested that differences between 4- and 8-cm mussels are generally on the order of 1.5°C or less, depending on wind

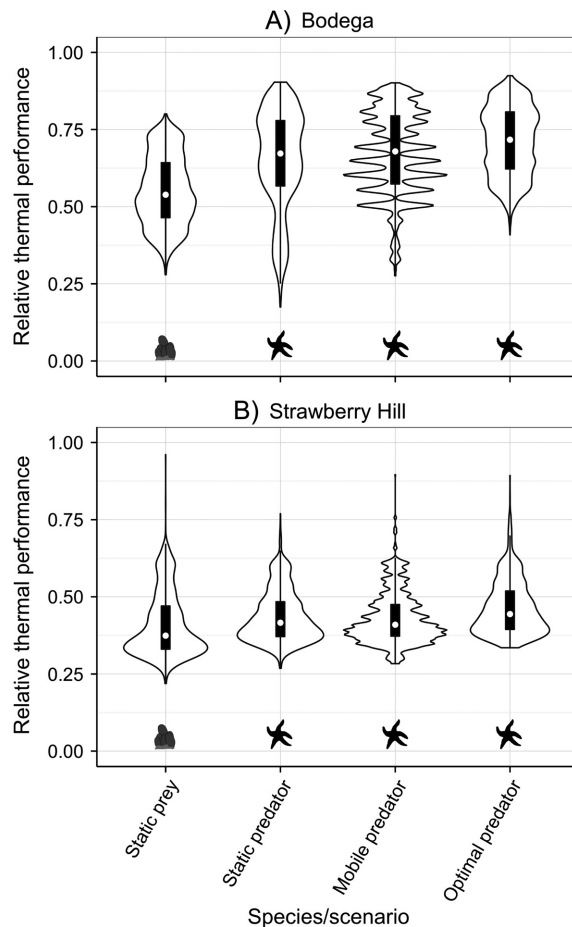


FIG. 4. Thermal performance calculated for *Mytilus californianus* (prey) and *Pisaster ochraceus* (predator) using species- and site-specific thermal performance curves and body temperatures recorded in situ by biomimetic sensors. The static scenarios represent individuals that remain immobile. The mobile predator considers movements between microhabitats, as informed by regular surveys of microhabitat use. The optimal scenario represents the hypothetical situation where *Pisaster* occupies the microhabitat that reports the highest performance. Panels A and B show data from Bodega and Strawberry Hill, respectively. Plot components are as in Fig. 1.

speed. Further investigations of the interplay between body size, microhabitat use, and physiological performance of ectotherms would be an interesting research avenue that can improve our mechanistic understanding of individual- and population-level responses to environmental change (Potter et al. 2013, Monaco et al. 2015).

Role of physiology

The thermal performance curves for *Pisaster* and *Mytilus* provide a means of quantifying mean levels of relative performance during continuous periods of high and low tides. While previous work had described aquatic thermal performance curves for *Pisaster* (Monaco et al. 2014) and *Mytilus* (Kroeker et al. 2014, Matzelle et al. 2015), aerial thermal performance curves had not been

described for either species. Given that both species spend significant amounts of time exposed at low tides, during which body temperatures fluctuate even more than at high tides (Elvin and Gonor 1979, Hofmann and Somero 1995), understanding the relationship between aerial body temperature and physiological performance is especially important. Note, however, that we had limited data to fit the aerial curve, so further efforts and especially further empirical measurements are warranted. Moreover, it is important to note that we based our metrics of thermal performance on respiration rate, and it is possible that results would have differed had we used a different metric of performance (Stoffels et al. 2016), for example foraging rate, which appears to have a lower thermal optimum than does respiration (Pincebourde et al. 2008). Integrating different metrics of performance that characterize whole organism physiological and ecological condition, is perhaps the best approach (Huey and Stevenson 1979, Stevenson et al. 1985, Huey and Kingsolver 1989).

A wealth of empirical and theoretical evidence suggests that the thermal ecology and physiology of ectothermic organisms are aligned (Angilletta 2009), although concerns have been raised that under global climate change such associations may be breaking down. For instance, Freitas et al. (2007) showed that the thermal sensitivity (i.e., T_A) and higher critical temperatures of various ectothermic species found in the Dutch Wadden Sea depend on the thermal environment where they are found. The same concepts apply for species distributions along larger (e.g., geographical) and smaller (e.g., vertical intertidal gradients) spatial scales (Stillman and Somero 2000, Monaco et al. 2010, Zippay and Hofmann 2010), all of which suggest that organisms are likely never “perfectly acclimated” at every location where they are found. Different models have been put forward to explain the links between body temperature variability and organisms’ thermal performance curve parameters (Gilchrist 1995, Vasseur et al. 2014, Dowd et al. 2015, Hong and Shurin 2015). Given the differences in body temperatures regularly experienced by *Pisaster* and *Mytilus* that result from both behavioral thermoregulation and differences in morphology and surface wetness, we expected that their thermal sensitivity parameters would differ too.

Quantifying patterns of body temperature in the field and comparing them against thermal sensitivity provides insight into how intertidal organisms contend with high spatial and temporal heterogeneity. Displaying a low T_A may offer a physiological buffer against potentially stressful extreme temperatures that might be reached in a more thermally heterogeneous environment (Hochachka and Somero 2002, Vasseur et al. 2014, Dowd et al. 2015, Hong and Shurin 2015), such as Strawberry Hill (Fig. 1), although such a strategy may reduce average fitness. It has also been suggested that a reduction of the temperature dependence of metabolism (i.e., lower T_A) is an energy conserving strategy for intertidal organisms

TABLE 3. Statistical results from multiple comparisons between relative thermal performances exhibited by groups of species vs. scenarios, as illustrated in Fig. 4.

Comparison	RMSE	Kendall's W		
		W	χ^2	P
Bodega				
Prey vs. static	0.16	0.77	3604.4	<0.0001
Prey vs. mobile	0.19 $\pm$ 0.0001	0.62 $\pm$ 0.0005	2884.6 $\pm$ 2.4	<0.0001
Prey vs. optimal	0.19	0.75	3487.1	<0.0001
Static vs. mobile	0.12 $\pm$ 0.0001	0.76 $\pm$ 0.0005	3538.1 $\pm$ 2.7	<0.0001
Static vs. optimal	0.13	0.90	4183.1	<0.0001
Mobile vs. optimal	0.06 $\pm$ 0.0002	0.80 $\pm$ 0.0007	3723.3 $\pm$ 3.14	<0.0001
Strawberry Hill				
Prey vs. static	0.09	0.72	3324.8	<0.0001
Prey vs. mobile	0.10 $\pm$ 0.00008	0.65 $\pm$ 0.0007	2979.9 $\pm$ 3.3	<0.0001
Prey vs. optimal	0.09	0.70	3232.9	<0.0001
Static vs. mobile	0.09 $\pm$ 0.0001	0.74 $\pm$ 0.0007	3420.4 $\pm$ 3.1	<0.0001
Static vs. optimal	0.07	0.77	3572.9	<0.0001
Mobile vs. optimal	0.09 $\pm$ 0.00008	0.75 $\pm$ 0.0008	3447.1 $\pm$ 3.9	<0.0001

Notes: Comparisons were made using root mean square error (RMSE) and the nonparametric test Kendall's W . Comparisons that included the mobile predator involved 500 independent contrasts (one for each simulated mobile individual), in which cases mean $\pm$ SE is shown. Results are provided for both sites, Bodega and Strawberry Hill.

(Marshall and McQuaid 2011; but see Huang et al. 2015). Thus, based on homeostatic considerations, lower T_A s were expected for species and sites showing the highest temperature variability, which conforms with what we observed for *Pisaster* (Table 2). Although *Mytilus* T_A was also higher at the more thermally variable site, Strawberry Hill, the site effect was not significant, and it is possible that more replication would have revealed significant differences. It is also conceivable that *Mytilus* relies more on other physiological mechanisms to cope with thermal variability; for example, production of Hsp70 or through metabolic depression via anaerobic metabolism (Anestis et al. 2007). Note that our estimates of T_A for *M. californianus* were lower than those reported by Van der Veer et al. (2006) for *M. edulis* ($T_A = 7,022 \pm 551$). This may be explained in part because *M. edulis* can more frequently be found in purely subtidal environments, and is therefore presumably adapted to more homogeneous conditions, and in part due to differences in local thermal conditions experienced by the individuals subjected to the experiments.

Although biomimetic loggers showed that temperatures were more variable at Strawberry Hill than Bodega (Fig. 1, Table 1), the thermal performance exhibited by organisms from both species showed higher variance at the latter site (Fig. 4). The lower T_A values estimated for the animals from Strawberry Hill may be a response to that higher temperature variability. Reaching extreme high and low levels of performance can be associated with a higher risk of experiencing critical conditions (Table 1), which appears to be ameliorated through a lower thermal sensitivity at Strawberry Hill. While these physiological responses, together with behavioral adaptations like selecting sheltered microhabitats (Monaco et al. 2015), seem to drive lower mean thermal performance at Strawberry Hill (Fig. 4), they reduce the risk of sporadic exposures to stressful temperatures. Thus, although both

Pisaster and *Mytilus* operate at suboptimal thermal performance levels, such strategy likely increases long-term fitness (Martin and Huey 2008).

When applying these performance curves to inter-species comparisons, our data showed that, regardless of scenario tested (static, mobile, or optimal), the thermal performance of the predator, *Pisaster*, was higher than that of its prey, *Mytilus* (Fig. 4, Table 3). These results support a prey stress model (PSM) variant within the environmental stress model framework, at least at these two sites. This finding contrasts with the results of Petes et al. (2008), presumably in part due to differences in methodology. Petes and colleagues directly assessed performance of individuals caged along an intertidal vertical (stress) gradient. Although informative, caging experiments may unnaturally influence the physiological condition of the organisms. It is known that *Pisaster* preferentially avoids exposure to the elements during low tides (Robles et al. 1995, Burnaford and Vasquez 2008), which could not occur in caging studies where the individuals were prevented from moving. Note that, traditionally, studies on ESMs have measured and compared performance at the same stress levels for different species, often along environmental gradients. Our approach differs in that we first consider the aspects of environmental drivers truly experienced by the organisms, in this case, their body temperatures, and then evaluated their condition.

Role of behavior

Our approach additionally allowed testing for the role of the predator's movement behavior in driving vulnerability (Sunday et al. 2014). Because *Pisaster* preferentially avoids exposure to wind and direct solar radiation during low tides (Robles et al. 1995, Burnaford and Vasquez 2008), we hypothesized that thermal performance would differ when microhabitat use was considered, as opposed



FIG. 5. Body mass indices (BMI) of (A) *Pisaster ochraceus* and (B) *Mytilus californianus* individuals collected at Bodega and Strawberry Hill during the summer of 2012. P values obtained from Welch's t tests indicate whether differences between sites were significant. Bars represent mean $\pm$ SE.

to static individuals that stayed fixed in the mid intertidal. Interestingly, at both sites evaluated, static predators performed almost as well as mobile predators (Fig. 4, Table 3). *Pisaster* has the ability to incorporate seawater into its coelomic cavity at high tide, and use it to increase thermal inertia at low tide (Pincebourde et al. 2009). Since our robo-sea stars are made of sponges that soak up water too, we were able to account for this process. This high thermal inertia in *Pisaster* may further explain the lack of differences we found between static and mobile predators. In particular, even during low tide, body temperatures recorded by robo-sea stars in sun-exposed environments were still quite frequently cooler than the energetic optimum predicted from TPCs. Thus, behavior offered no additional opportunity to increase performance, except during rare but extreme conditions when exposed body temperature was higher than the optimum. While unexpected, our observations are supported by calculations from Fly et al. (2012), who found very modest differences in thermal energy costs for individuals located at different shore levels. These accounts, however, do not conform with the analysis from previous studies evaluating the effects of prey addition/removal on the intertidal distribution of *Pisaster* (Robles et al. 1995, Robles and Desharnais 2002). Their experiments showed that *Pisaster* could move vertically on the shore depending on availability of profitable prey items. However, because sea stars would remain in the low intertidal zone when food was available, the authors argued that reaching higher shore levels would imply additional energy costs to cope with thermal stress, which our data did not detect.

Although we found only marginal differences in performance between mobile and static predators (Fig. 4,

Table 3), this does not mean they did not exist. Our method only accounted for the thermal response, ignoring other relevant variables that have been demonstrated to influence behavior and potentially fitness in *Pisaster*, most noticeably, desiccation (Feder 1956, Landenberger 1969) and avoidance of predation by gulls (Rogers and Elliott 2013). Since *Mytilus* inhabits higher shore levels than *Pisaster*, it is potentially better adapted to cope with factors such as desiccation. The differential effect of these factors on this predator–prey interaction, and not just body temperature, may help explain the results by Petes et al. (2008), which led them to define this system as consumer stress model. As our study demonstrates, however, simply knowing an organism's critical or optimal temperature alone is not sufficient to gauge relative stress. Mussels consistently exhibited higher CT_{max} values than their predators (Table 2), but nevertheless had a smaller safety margin at Strawberry Hill (Table 1), because, even under identical microclimates, they had higher body temperatures (Broitman et al. 2009, Monaco and Helmuth 2011). It is also notable that patterns of relative stress based on average performance were, in some cases, at odds with predictions based on extremes. At Bodega, for example, sea stars had, on average, higher performance than mussels, but also had a slightly smaller thermal safety margin (Table 1). At Strawberry Hill, in contrast, the predator experienced both higher average performance as well as a substantially greater thermal safety margin.

The optimal predator scenario data provided an interesting viewpoint of the system. Although its thermal performance was higher than both the static and mobile predator scenarios, the differences were nominal (Fig. 4, Table 3). Thus, among those options available for *Pisaster*,

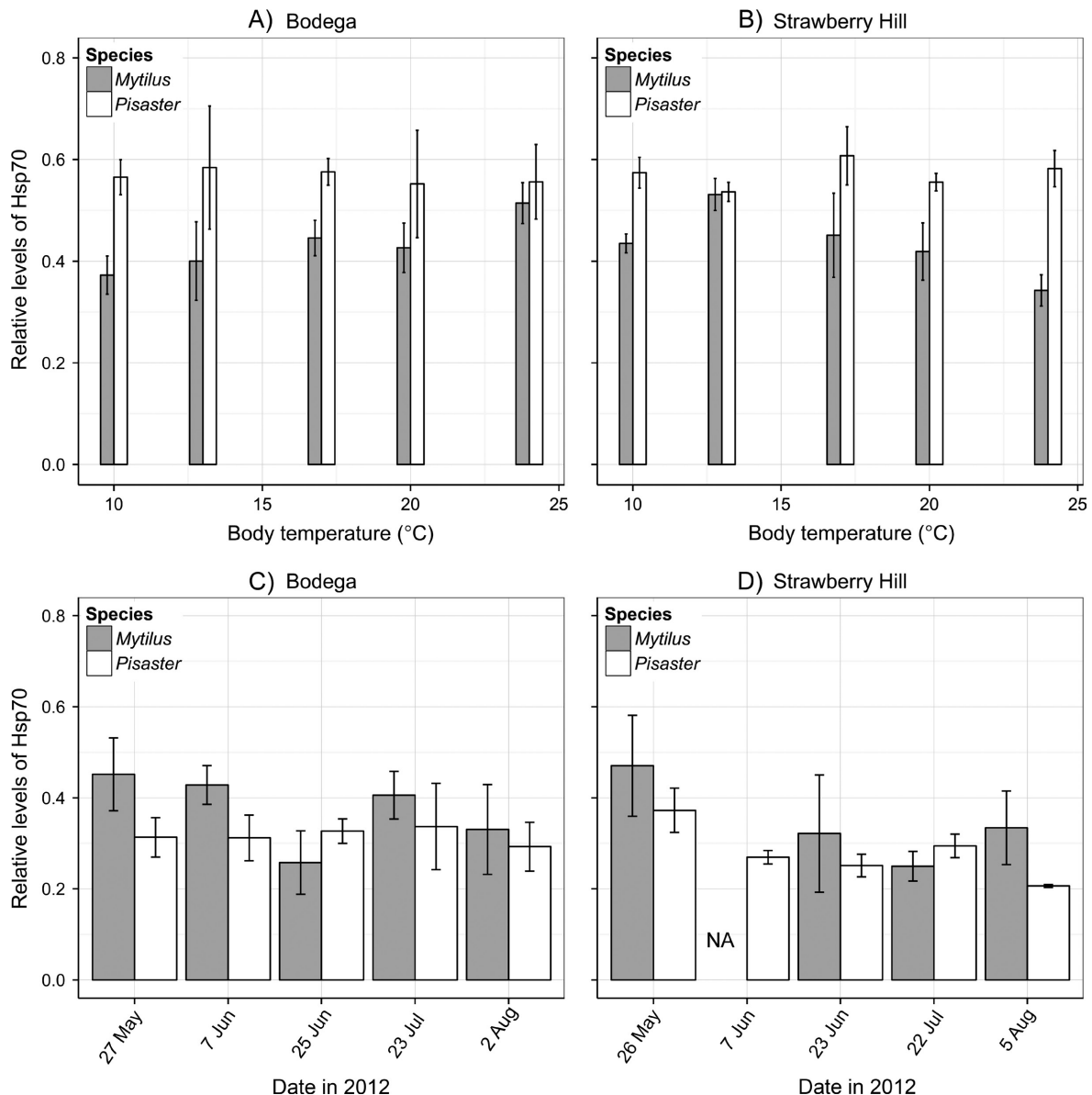


FIG. 6. Total heat shock protein 70 kDa (Hsp70) expression (i.e., constitutive and inducible isoforms are not differentiated) measured from *Pisaster ochraceus* tube feet and *Mytilus californianus* gill tissues. In panels A and B, tissues were sampled after individuals had been kept at one of six seawater temperature treatments (10°, 13°, 18°, 21°, 24°, and 27°C) for 4 d. In panels C and D, tissues were collected in the field at low tide and immediately frozen in dry ice. Bars represent mean $\pm$ SE.

none represented a strikingly advantageous one. On the contrary, based on the risk of temperature-related mortality (i.e., TSM), an optimal scenario may not be as beneficial as expected. For instance, at Strawberry Hill an increase in relative performance would have placed *Pisaster* closer to its critical temperature during rare extremes (Table 1; Vasseur et al. 2014, Dowd et al. 2015, Hong and Shurin 2015). Including higher shore levels in the analyses may alter results; however, we deliberately ignored *Pisaster* thermal performance at higher shore levels and focused on the zone where their interaction with *Mytilus* is strongest.

Comparably, although our estimate of overall relative performance ignored activity during neap tides, future studies using these or other species could explicitly account for rhythmicity in important behaviors that contribute to fitness directly (e.g., foraging, Hayford et al. 2015) or indirectly (e.g., non-consumptive effects; Matassa and Trussell 2015).

Empirical indicators of physiological condition

We accompanied our indirect metric of thermal performance with direct, empirically determined indicators of

physiological condition, BMI and Hsp70. Because these were measured from field-collected individuals, they potentially represent the net effect of biotic and abiotic factors on the surveyed populations, as well as the influence of behavior on thermal performance. Although these are not directly comparable to our estimates based on thermal performance curves, they could be correlated.

Both indicators BMI and Hsp70 have been used to examine physiological condition in sea stars and mussels. BMI, a ratio between soft tissues produced and total somatic mass (including calcareous components such as shells or skeletons), provides broad insights about the individual's general capacity to grow and reproduce. Only *Mytilus* BMI changed with site, being higher at Strawberry Hill (Fig. 5). The link with thermal performance, however, is not particularly clear given that mean performance at Strawberry Hill was lower (Fig. 4). BMI variability has been primarily associated with changes in temperature and food availability, as well as their interaction (Sanford and Menge 2007, Schneider et al. 2010, Fitzgerald-Dehoog et al. 2012). If body temperature were the main driver of *Mytilus* BMI, then we would have incorrectly predicted higher performance at Bodega based on the thermal performance estimates. This evidence suggests that some variable other than body temperature allowed for higher *Mytilus* BMI at Strawberry Hill, food availability (quantity, quality, proportion of time spent submerged filter-feeding) being the most likely candidate (Honkoop and Beukema 1997, Dahlhoff et al. 2001, Dowd et al. 2013). Because we collected mussels from approximately the same intertidal elevation at both sites, we assume that time foraging were similar between populations. To test whether food quantity varied, we compared levels of chlorophyll *a* ($\mu\text{g/L}$) determined via satellite imaging 1 km offshore from each site (data available online).⁴ Monthly measurements taken during the five months preceding the animal collections revealed no differences between the two sites. Thus, we have no evidence to suggest that food availability explained the pattern in BMI. Thus, food quality or diet composition, which we did not quantify, might explain our results.

As revealed by total Hsp70 expression data, we did not observe signs of sublethal thermal stress on the species and populations examined. First, our lab experiment showed no change in Hsp70 along an ecologically realistic range of body temperatures (Fig. 6A, B). This was expected given that the inducible isoform of Hsp70 is known to up-regulate in situations of more critical cellular damage (Feder and Hofmann 1999). The levels detected are most likely explained by the constitutive isoform, which is always present (Petes et al. 2008, Place et al. 2008). We also found no differences between sites, indicating that these populations do not differ in their baseline heat shock response. This lack of response in Hsp70 does not conform to the more sensitive SMR response detected at the same treatment temperatures

(Fig. 2). Our lab experiment does not indicate whether populations would respond differently under more stressful scenarios, which are possible based on Fig. 1. According to the higher variability in thermal performance observed at Bodega, one might expect an increased heat shock response there, but this may vary between species. While both species are certainly capable of mounting a heat shock response, evidence suggests that *Mytilus* up-regulates Hsp70 more readily and strongly than *Pisaster* (Halpin et al. 2004, Petes et al. 2008). This is expected given that *Pisaster* can rely on movement between microhabitats (Huey 1991) and its high thermal inertia (Pincebourde et al. 2009).

It may be argued that our short acclimation (5 d) and treatment (4 d) periods would not clear any endogenous rhythm in *Pisaster* associated to the two-week tidal cycles, which would be reflected by the physiological performance metrics measured (Hsp70 and SMR). While this is a possibility that warrants further investigation, evidence also suggests that these traits do not follow endogenous tidal rhythms in all intertidal animals (Todgham et al. 2006, Schill et al. 2008, Connor and Gracey 2011).

Second, our field-collected individuals showed no signs of thermal stress both over time and between sites (Fig. 6C, D). Interestingly, the *Mytilus* response was more variable than that of *Pisaster*, which is possibly a consequence of the higher temporal variability in body temperature and the prey's ability to mount a heat shock response quicker (Halpin et al. 2004, Petes et al. 2008). Similarly, the variability in Hsp70 production was greater at Strawberry Hill than Bodega, which may be due to the higher temporal variability in biomimetic temperature observed there (Fig. 1).

It should also be pointed that we only measured Hsp70 levels on single tissue types for both species, gills in *Mytilus* and tube feet in *Pisaster*. We chose these because they had been successfully utilized previously (Petes et al. 2008), and because their integrity is critical for the animals' functioning. Although we ensured sampling tissues from multiple areas in the body, it is unclear whether other organs might exhibit different heat shock responses (Dahlhoff 2004).

Overall, as empirical indicators of physiological performance, BMI and Hsp70 loosely corroborated the insights gathered through our observational-modeling approach. Because BMI and Hsp70 depend on other intrinsic (e.g., phenotypic plasticity) and extrinsic (e.g., physical forces, food availability) processes besides temperature, which interact in complex ways, caution must be taken when drawing direct connections between them.

An important caveat to the way we applied our approach here is that we ignored potential effects of additional relevant environmental parameters on individual's performance. Follow-ons to this study may incorporate these elements. The impact of multiple environmental stressors, for example, has recently regained popularity in the marine ecology literature, especially given the

⁴ data.cencoos.org

urgency imposed by ongoing climate change (Harley et al. 2006). In particular, nutrition may condition organisms' thermal performance response (Freitas et al. 2007, Fitzgerald-Dehoog et al. 2012), which could be considered by monitoring food availability for both predators and prey. And, as recently discovered, temperature also affects the prevalence of sea star wasting disease (Bates et al. 2009, Eisenlord et al. 2016).

Our results may have implications for future climate change scenarios, depending on site. Higher T_A values for the predator, as observed in Bodega, imply that future warmer scenarios will increase their performance more than that of the prey. A comparable analysis Freitas et al. (2007) found similar results, in that prey showed lower thermal sensitivity than their consumers. We found the opposite at Strawberry Hill, where T_A was higher for *Mytilus*; thus warmer conditions may increase performance of the prey more than that of the predator, potentially releasing some predation pressure, as long as temperatures do not exceed CT_{max} . Given the keystone role of *Pisaster* (Paine 1966, 1974), ecological implications of this prediction may reach the community level.

Conclusions

This study provides a novel approach for applying the ESM framework based on species' traits (behavior, physiology) that may not be detectable using more traditional experimental field techniques. Using a combination of field observations and modeling, we were able to quantify thermal performance of the predator *Pisaster* and its main prey *Mytilus*. Applying this approach revealed that the thermal performance of the predator is higher than that of the prey at these two sites (i.e., PSM). Both species appear to buffer against thermal heterogeneity, characteristic of their intertidal environment, by reducing thermal sensitivity. Thus, the role of physiology appears important in driving responses in this predator–prey system, and most likely others where environmental variability is as conspicuous as the rocky shore. Our data lend support to the idea that both average conditions as well and variance in those conditions can determine the importance of environmental drivers (Vasseur et al. 2014). Movement behavior did not appear related to average thermal performance, although it may have reduced the incidence of rare extreme events (Monaco et al. 2015). At least for the period considered, the physiological cost of thermal stress did not drive the predator's microhabitat use at either site. Its movement behavior may be driven more by alternative environmental forces such as desiccation, wave action, predator avoidance, or as a bet-hedging strategy against extremes.

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DATA AVAILABILITY

Data associated with this paper are available in Dryad: <http://dx.doi.org/10.5061/dryad.30hg1>